

Question - 01

```
03 -- zsh
sharjeel < 03 >: run 01-text-editor
===== TEXT EDITOR UNDO/REDO SYSTEM =====

Action Performed: Insert "H" at position 0
Current Document: H
Action Performed: Insert "e" at position 1
Current Document: He
Action Performed: Insert "l" at position 2
Current Document: Hel
Action Performed: Insert "l" at position 3
Current Document: Hell
Action Performed: Insert "o" at position 4
Current Document: Hello

----- ACTION HISTORY -----
1. Insert "H" at position 0
2. Insert "e" at position 1
3. Insert "l" at position 2
4. Insert "l" at position 3
5. Insert "o" at position 4
-----

Action Performed: Delete "o" from position 4
Current Document: Hell

Undo Operation Performed
Reverted: Insert "o" at position 4
Current Document: Hello
Undo Operation Performed
Reverted: Delete "o" from position 4
Current Document: Hell
Redo Operation Performed
Reapplied: Insert "o" at position 4
Current Document: Hello

Adding New Action:
Action Performed: Insert "!" at position 5
Current Document: Hello!

----- UPDATED HISTORY -----
1. Insert "H" at position 0
2. Insert "e" at position 1
3. Insert "l" at position 2
4. Insert "l" at position 3
5. Insert "o" at position 4
6. Delete "o" from position 4
7. Insert "!" at position 5
-----

Applying History Limit: 4 actions
Oldest actions removed successfully.
----- HISTORY AFTER TRUNCATION -----
1. Insert "l" at position 3
2. Insert "o" at position 4
3. Delete "o" from position 4
4. Insert "!" at position 5
-----

Current Document State: Hello!

===== PROGRAM TERMINATED =====
```

Question - 02

```
03 — zsh
sharjeel < 03 >: run 02-maze-solver
Solution found!
* * # . .
# * # . #
. * * * *
. # # # *
* * * * *
* D . . .

Total Recursive calls: 29
(0, 0), (0, 1), (1, 1), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 0), (4, 1), (4, 2), (4, 3), (4, 4), (5, 0)
sharjeel < 03 >: █
```